Question	Answer	Marks
3(a)	$\omega^2 = 2g / L$	**C1**
	$T = 2\pi / \omega$	**C1**
	$\omega^2 = (2 \times 9.81) / 0.19$ $\omega = 10.2$ (rad s^{-1}) $T = 2\pi / 10.2$ $= 0.62$ s	**A1**
3(b)(i)	e.g. viscosity of liquid/friction within the liquid/viscous drag/friction between walls of tube and liquid	**B1**
3(b)(ii)	(maximum) KE = $½mv_0^2$ **and** $v_0 = \omega x_0$ **or** energy = $½m\omega^2 x_0^2$	**C1**
	change $= ½ \times 18 \times 10^{-3} \times 103 \times [(2.0 \times 10^{-2})^2 - (0.95 \times 10^{-2})^2]$	**C1**
	$= 2.9 \times 10^{-4}$ J	**A1**

Question	Answer	Marks
4(a)	pulses (of ultrasound from generator)	B1